

BME 313L: Introduction to Numerical Methods in Biomedical EngineeringHonor Code: *I have neither given nor received aid on this quiz.*

UT EID: _____

Printed Name: _____

Signature: _____

Problem 1 (5 points): For each statement, state whether it describes model **qualification**, **verification**, or **validation**.

- (a) A group building a pharmacokinetic model of drug clearance asks whether treating the body as a single well-mixed compartment is defensible for a drug that accumulates in fat.

ANSWER (1 pt): Qualification. The question is whether the modeling assumption suits the intended use, and it is asked before any code exists.

- (b) A student implements a new ODE solver for the part (a) compartment model. She runs it on $y' = -2y$, whose exact solution she knows, confirming that the code recovers the known solution.

ANSWER (1 pt): Verification. She is checking that the implementation solves the equations she chose, using a problem whose exact solution is known.

- (c) A lab predicts peak wall stress in an aneurysm from a patient's CT scan and compares the prediction against strain-gauge readings taken from the excised tissue.

ANSWER (1 pt): Validation. The prediction is compared against an independent physical measurement of the real system.

- (d) In one sentence, explain why passing (b) tells you nothing about whether (a) is satisfied.

ANSWER (2 pts): Verification only shows that the equations were solved correctly. A correctly solved but wrong model is still wrong, so passing (b) says nothing about whether the well-mixed compartment assumption is appropriate for a drug that accumulates in fat.

Problem 2 (5 Points): (a) Write down the contents of `z`. (b) In one sentence, what mathematical operation does the script implement?

```
x = [0 -3 4 6];
y = [-1 2 4 -4];
z = zeros(4,1);
disp('Problem 2')
```

```
for i = 1:4
    z(i) = y(i);
    if( x(i) < y(i) )
        z(i) = x(i);
    end
end
```

```
end
```

ANSWER (a) (3 pts): $z = [-1; -3; 4; -4]$ (a 4x1 column vector)

ANSWER (b) (2 pts): The script computes the element-wise minimum of x and y , i.e. $z = \min(x,y)$.

Problem 3 (5 Points): The code below will choose and display what the kids watch on TV this evening. What is the probability that we are watching K-Pop Demon Hunters?

```
1 r = rand;
2 if(r > -1.0)
3     disp('K-Pop Demon Hunters')
4 else
5     disp('Jurassic World: Camp Cretaceous')
6 end
```

A) Probability: _____

B) Re-write line 2 so that the probability of watching Jurassic World is 55%: _____

ANSWER A): 1. `rand` returns a value in the interval (0,1), so $r > -1.0$ always holds and the if branch executes every time. K-Pop Demon Hunters is displayed with certainty.

ANSWER B): `if(r > 0.55)`

Equally correct: `if(r >= 0.55)`, because `rand` is continuous, or `if(r < 0.45)` with the two titles swapped between the branches.

Problem 4 (15 Points): The *random Fibonacci sequence* is defined by

$$x_1 = x_2 = 1, x_{k+1} = x_k \pm x_{k-1},$$

where at each step the sign is chosen at random, resulting in plus or minus, with equal probability, independently of every other step. Write a MATLAB function `randFibVals(n)` that returns the first n terms of a random Fibonacci sequence as a column vector.

ANSWER (15 pts):

```
function x = randFibVals(n)
% Return the first n random Fibonacci numbers as a column vector.
x = zeros(n,1);
x(1) = 1;
x(2) = 1;
for k = 3:n
    if( rand < 0.5 )
        x(k) = x(k-1) - x(k-2);
    else
        x(k) = x(k-1) + x(k-2);
    end
end
```

Also fully correct: $x(k) = x(k-1) + \text{sign}(\text{rand}-0.5)*x(k-2)$, or drawing the sign as $s = 2*\text{randi}([0 \ 1]) - 1$.